TORQUE CALCULATION AND MOTOR SELECTION

NEMA 23 torque= 1.2-3.0N.m

MG90s Metal-Gear Servo torque=~0.22N.m

MG90 micro servo torque =~0.18 N.m

MG996R torque =0.92-1.08N.m

Total torque=mass*acceleration*distance

Link 1 center of mass assumed at 0.07 m

Link 2 center of mass assumed at 0.195 m

Link 3 center of mass assumed at 0.25 m

Link 4 center of mass assumed at 0.275 m

$T1=(0.20*9.81*0.07)+(0.15*9.81*0.195)+(0.28*9.81*0.25)+(0.10*9.81*0.275)$

$= 0.13734+0.2869425+0.6867+0.269775=1.3808Nm$ (joint 1 :base)

NEMA 23(30-60 RPM) is used.

$T2=(0.15*9.81*0.055)+(0.28*9.81*0.11)+(0.10*9.81*0.135)$

$=0.0809325+0.302148+0.132435=0.5156Nm$ (joint 2:elbow)

MG996R(40-50 RPM) is used.

$F3=(0.28+0.10)*9.81=3.73N$

$T3=3.75*0.008$(lead screw)/ $2pi*0.35=0.0135Nm$ (joint3:Z/vertical)

MG90s Metal-Gear Servo(10-20 RPM) is used.

$T4<0.05N$, (joint 4:wrist)

MG90 micro servo (80-100 RPM) is used